36. Containers, Namespaces, and Networking

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Phase: Phase 11 — Containers and cloud Linux
Estimated time: 2 days
Slug: `36-docker-linux-net`

Learning objectives

- Name major namespaces (pid, net, mnt, user)
- Explain bridge networking basics
- Debug DNS inside containers
- Use host networking knowingly

Prerequisites

Complete prior numbered lessons. Prefer a real Linux VM/container for Linux-specific topics.

Linux containers = namespaces + cgroups + union filesystems.

```
docker network ls
docker run --rm busybox nslookup example.com
```

Network modes: bridge (default), host, none, overlay (swarm/k8s).

When networking “feels broken”, check: DNS, published ports, firewall, and whether you expected localhost on host vs container.

Common mistakes

- –network host as default habit
- Assuming localhost in container is the host
- Ignoring cgroup memory limits

Exercises

Work in `exercises/36-docker-linux-net/`. Run:

```
./test.sh
```

- **A (Drill):** make `./test.sh` pass
- **B (Apply):** note one real command you ran on a Linux host related to this lesson
- **C (Challenge):** rewrite your solution from memory the next day

Do not open `solutions/36-docker-linux-net/` until you have an honest failing attempt.

Self-check

1. What do namespaces isolate?
2. Default docker network mode?

3. localhost confusion?

Further reading

- https://docs.docker.com/network/